Potential of chopped heath biomass and spent growth media to replace wood chips as bulking agent for composting high N-containing residues

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Highlights

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Chopped heath and spent growth media can be used as bulking agents.

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Chopped heath compost has a high OM content and C/P ratio.

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Clear differences in N immobilization between bulking agents were observed.

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NIRS can monitor quickly the chemical properties and stability during composting.

Abstract

We investigated the potential of C-rich byproducts to replace wood chips as bulking agent (BA) during composting. The impact of these alternatives on the composting process and on compost stability and characteristics was assessed. Three BA (chopped heath biomass and spent growth media used in strawberry and tomato cultivation) were used for processing leek residues in windrow composting. All BA resulted in stable composts with an organic matter (OM) content suitable for use as soil amendment. Using chopped heath biomass led to high pile temperatures and OM degradation and a nutrient-poor compost with high C/P ratio appropriate for increasing soil organic carbon content in P-rich soils. Spent substrates can replace wood chips, however, due to their dense structure and lower biodegradation potential, adding a more coarse BA is required. Generally, the nutrient content of the composts with growth media was higher than the composts with wood chips and chopped heath biomass.

Graphical abstract

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Introduction

Plant pathogens, phytotoxicity, variable nutrient status and a high pH and electrical conductivity (EC) in compost can result in a variable product quality (Noble and Roberts, 2004). In order to ensure a good composting process, i.e., the microbial conversion of fresh organic material into a stable and humus-rich product, feedstock materials characterized by a low dry matter (DM) content and a low C/N ratio should be co-composted with drier, C-rich bulking agents (BA) to provide structure and porosity to the mixture to ensure proper aeration (Barrington et al., 2002), decrease the moisture content and supply biodegradable C. Sufficient C levels lead to microbial immobilization of N during the composting process, thereby minimizing N losses (Nolan et al., 2011) as well as generating enough metabolic heat to reach thermophilic temperatures and sanitize the compost. Commonly-used BA such as wood chips or tree bark are well-known and well-studied (e.g. Larney et al., 2008, Suzuki et al., 2004, Vandecasteele et al., 2013), but these wood-based BA may become more scarce and more expensive, causing a barrier to on-farm composting (Viaene et al., 2016) and they can contain high concentrations of heavy metals (Vandecasteele et al., 2013).

Some studies, mainly from Mediterranean regions, have tested other byproducts for use as BA, including sugar cane bagasse (Meunchang et al., 2005), rice hulls (Leconte et al., 2009) and brans (Chang and Chen, 2010), grape stalks (Cayuela et al., 2010), olive leaves (Alburquerque et al., 2006), sawdust (Huang et al., 2004), biochar (Dias et al., 2010), paper-cardboard (Francou et al., 2008) and straw (Michel et al., 2004) for co-composting kitchen waste and various types of manure. However, byproducts from nature management and greenhouse cultivation have been rarely tested as alternative BA. A potential byproduct from nature management is chopped heath biomass, obtained by a less intensive sod-cutting procedure for removing excess nutrients in heathland management. This process is gaining more interest as it has several advantages over standard sod-cutting (Niemeyer et al., 2007). The chopped biomass consists of above-ground vegetation and a large part of the O-horizon(s). Approximately 87 t fresh material ha^{-1} can be annually harvested (Viaene et al., 2014). In greenhouse cultivation, substrate reutilization is strongly encouraged because the disposal of growth media at the end of the growing season is a potential threat to the environment (Diara et al., 2012). Spent substrates (annually ca. 35 t ha^{-1}, Viaene et al. (2014)) can either be recycled as a soil amendment or mixed with other substrates after composting (Diara et al., 2012).

The main objective of this study is to test C-rich byproducts for their value as BA with the aim of replacing wood chips for co-composting N-rich vegetable crop residues. N-rich vegetable crop residues, such as leek residues generated during the cleaning process of leek, pose a high risk for N losses when applied to or left on the field during autumn (Chaves et al., 2007). Management of those crop residues is an effective strategy to reduce N losses, notably nitrate leaching and probably N_2O emissions (de Ruijter et al., 2010). On-farm composting of vegetable residues is a valuable management option. Not only would it reduce N losses, it promotes the local recycling of nutrients and organic matter (OM) and the production of a soil-improving compost (Viaene et al., 2017). We tested three locally available alternative BA, namely chopped heath biomass and spent strawberry and tomato substrates. The performance of each BA was evaluated for its ability to ensure an optimal composting process (OM degradation, sufficiently high N immobilization and sufficiently high pile temperatures for hygienic reasons) resulting in a mature compost (microbial and biochemical stability, low N immobilization) with soil improving characteristics (pH, EC, bulk density, OM and nutrient content, N fertilizer replacement value). To the best of our knowledge, this is the first study on composting of chopped heath biomass or spent substrates from greenhouse cultivation and the evolution of N immobilization capacity during the composting process. A second (methodological) objective of this study is to assess whether the critical parameters of the composting process and compost quality related to chemical composition and stability could be monitored by near infrared spectroscopy (NIRS). Characterization of chemical properties and stability of feedstock materials and end products of composting allows for better process control and focused compost application, but lab-based analyses are time-consuming and thus expensive. NIRS has been reported to allow for fast screening of chemical and biochemical properties of compost (Galvez-Sola et al., 2010a, Galvez-Sola et al., 2010b) and soil (Terhoeven-Urselmans et al., 2008).

Composting process

The compost trial was executed in open air on a concrete pad at the composting facility of the experimental farm of the Institute for Agricultural and Fisheries Research (ILVO) in Merelbeke, Belgium. On January 16th, 2014 four windrows were set up; in the beginning of the process, each windrow was 12 m long, 3 m wide and 1.5 m high. Each windrow consisted of 200 kg of fresh straw and 4225 kg fresh weight (FW) of leek crop residues, collected at the farm after harvested leek was cleaned and

Feedstock characterization: bulking agents, leek residues and mixtures

The two batches of leek residues added during the composting process had a very similar composition (Table 1). The BA were characterized by a higher OM and DM content, lower total N content and consequently higher C/N ratio than the leek residues (Table 1). In comparison to wood chips and straw, the alternative BAs had a lower OM content, higher N content and lower C/N ratio (Table 1). Chopped heath biomass had the lowest OM content (72.6% of DM), followed by tomato substrate (78.6% of DM) and

Composting process: temperature and CO_2 patterns

After a first temperature peak, pile temperatures of Cw suddenly decreased to 14 °C after 12 days, while enough oxygen was available. Five hypotheses could explain why the composting process slowed down for Cw: (1) A lack of easily available N. This seems unlikely, as adding extra N via 2250 kg fresh leek on day 19 did not cause pile temperatures to rise, consequently N is certainly not limiting (Fig. 1A). (2) A lack of easily available C. Because almost all N became immobilized after adding

Conclusions

Chopped heath biomass and both types of spent growth media can be used as BA for co-composting green crop residues, but they resulted in different composting processes and end products. More specifically, when using spent growth substrates, the compost requires more frequent turning in comparison with wood chips due to oxygen shortage in these dense mixtures. To prevent oxygen shortage when applying these alternative BA, a coarser BA (i.e., larger particle size) with more structure could be

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